PROBLEMS

Section 7.2—Biot–Savart's Law

7.1 (a) State Biot–Savart's law.

(b) The y- and z-axes, respectively, carry filamentary currents 10 A along $\mathbf{a}_y$ and 20 A along $-\mathbf{a}_z$. Find $\mathbf{H}$ at $(-3, 4, 5)$.

7.2 A current element $I\,d\mathbf{l} = 4\mathbf{a}_x$ A · m is located at the origin. Determine its contribution to the magnetic field intensity at (a) $(1, 0, 0)$, (b) $(0, 1, 0)$, (c) $(0, 0, 1)$, (d) $(1, 1, 1)$.

7.3 Two infinitely long wires, placed parallel to the z-axis, carry currents 10 A in opposite directions as shown in Figure 7.31. Find $\mathbf{H}$ at point P.

7.4 A current filament lies along the entire z-axis and carries 12 A in the $-\mathbf{a}_z$ direction. Determine $\mathbf{H}$ at $(3, 4, -1)$.

7.5 A conducting filament carries current I from point $A(0, 0, a)$ to point $B(0, 0, b)$. Show that at point $P(x, y, 0)$,

$$\mathbf{H} = \frac{I}{4\pi\sqrt{x^2+y^3}}\left[\frac{b}{\sqrt{x^2+y^2+b^2}} - \frac{a}{\sqrt{x^2+y^2+a^2}}\right]\mathbf{a}_\phi$$

7.6 Consider AB in Figure 7.32 as part of an electric circuit. Find $\mathbf{H}$ at the origin due to AB.

7.7 Line $x = 0$, $y = 0$, $0 \le z \le 10$ m carries current 2 A along $\mathbf{a}_z$. Calculate $\mathbf{H}$ at points

(a) $(5, 0, 0)$

(b) $(5, 5, 0)$

(c) $(5, 15, 0)$

(d) $(5, -15, 0)$

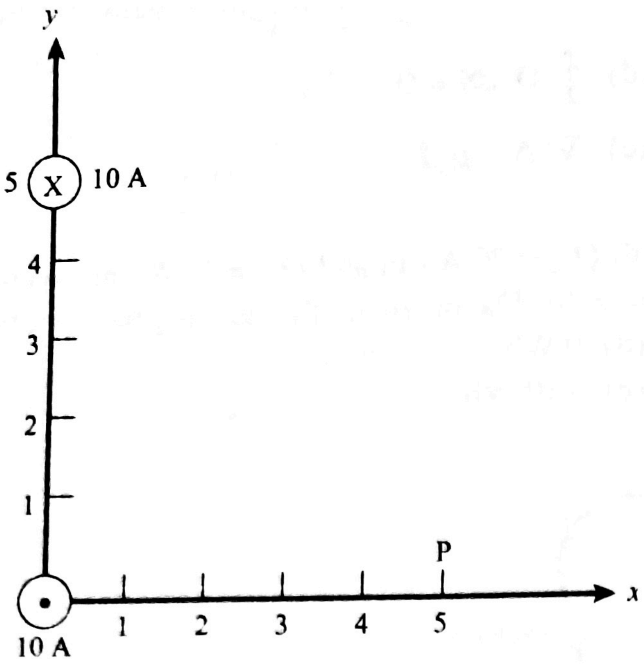

Figure 7.31 For Problem 7.3.

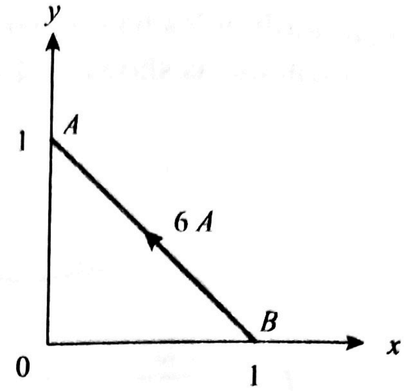

Figure 7.32 For Problem 7.6.

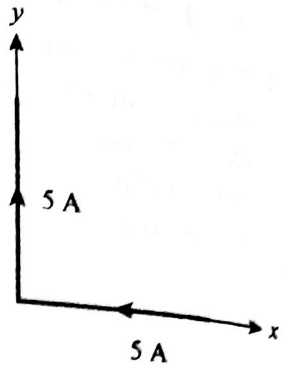

Figure 7.33 Current filament for Problem 7.9.

***7.8** (a) Find **H** at (0, 0, 5) due to side 2 of the triangular loop in Figure 7.6(a).
(b) Find **H** at (0, 0, 5) due to the entire loop.

7.9 An infinitely long conductor is bent into an L shape as shown in Figure 7.33. If a direct current of 5 A flows in the conductor, find the magnetic field intensity at (a) (2, 2, 0), (b) (0, −2, 0), and (c) (0, 0, 2).

7.10 A rectangular loop carrying 10 A of current is placed on $z = 0$ plane as shown in Figure 7.34. Evaluate **H** at
(a) (2, 2, 0)
(b) (4, 2, 0)
(c) (4, 8, 0)
(d) (0, 0, 2)

7.11 Find **H** at the center C of an equilateral triangular loop of side 4 m carrying 5 A of current as in Figure 7.35.

7.12 A square conducting loop of side 4 cm lies on the $z = 0$ plane and is centered at the origin. If it carries a current 5 mA in the counterclockwise direction, find **H** at the center of the loop.

***7.13** (a) A filamentary loop carrying current I is bent to assume the shape of a regular polygon of n sides. Show that at the center of the polygon

$$H = \frac{nI}{2\pi r}\sin\frac{\pi}{n}$$

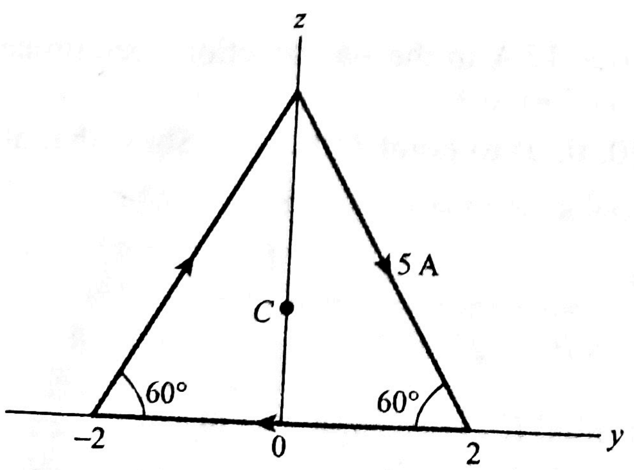

Figure 7.35 Equilateral triangular loop for Problem 7.11.

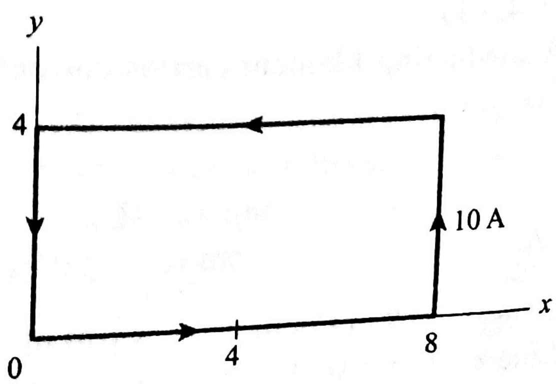

Figure 7.34 Rectangular loop of Problem 7.10.

where r is the radius of the circle circumscribed by the polygon.

(b) Apply this for the cases of $n = 3$ and $n = 4$ and see if your results agree with those for the triangular loop of Problem 7.10.

(c) As n becomes large, show that the result of part (a) becomes that of the circular loop of Example 7.3.

7.14 For the filamentary loop shown in Figure 7.36, find the magnetic field strength at O.

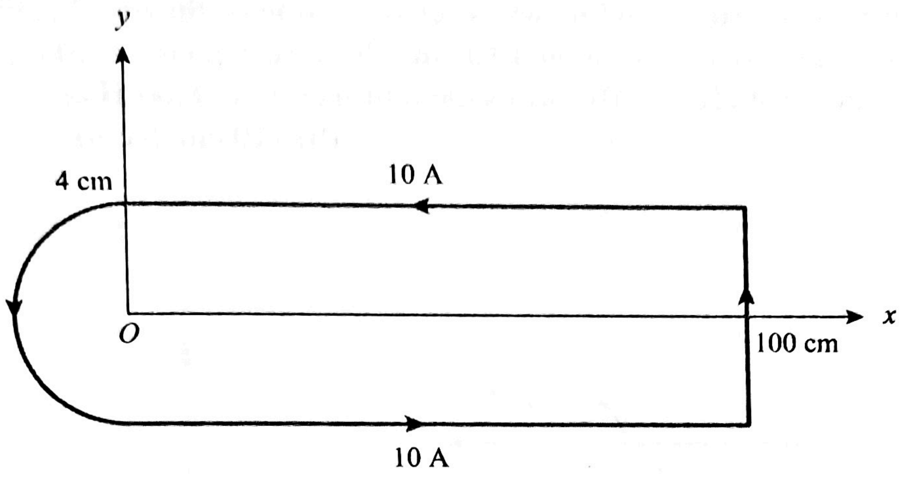

Figure 7.36 Filamentary loop of Problem 7.14 (not drawn to scale).

7.15 Two identical current loops have their centers at (0, 0, 0) and (0, 0, 4) and their axes the same as the z-axis (so that the "Helmholtz coil" is formed). If each loop has a radius of 2 m and carries a current of 5 A in $\mathbf{a}_\phi$, calculate $\mathbf{H}$ at

(a) (0, 0, 0) (b) (0, 0, 2)

7.16 A solenoid of radius 4 mm and length 2 cm has 150 turns/m and carries a current of 500 mA. Find (a) $|\mathbf{H}|$ at the center, (b) $|\mathbf{H}|$ at the ends of the solenoid.

7.17 Plane $x = 10$ carries a current of 100 mA/m along $\mathbf{a}_z$, while line $x = 1$, $y = -2$ carries a filamentary current of 20π mA along $\mathbf{a}_z$. Determine $\mathbf{H}$ at (4, 3, 2).

Section 7.3—Ampère's Circuit Law

7.18 (a) State Ampère's circuit law.

(b) A hollow conducting cylinder has inner radius a and outer radius b and carries current I along the positive z-direction. Find $\mathbf{H}$ everywhere.

7.19 Current sheets of $20\mathbf{a}_x$ A/m and $-20\mathbf{a}_x$ A/m are located at $y = 1$ and $y = -1$, respectively. Find $\mathbf{H}$ in region $-1 < y < 1$.

7.20 The $z = 0$ plane carries current $\mathbf{K} = 10\mathbf{a}_x$ A/m, while current filament situated at $y = 0$, $z = 6$ carries current I along $\mathbf{a}_x$. Find I such that $\mathbf{H}(0, 0, 3) = \mathbf{0}$.

7.21 (a) An infinitely long solid conductor of radius a is placed along the z-axis. If the conductor carries current I in the $+z$ direction, show that

$$\mathbf{H} = \frac{I\rho}{2\pi a^2}\mathbf{a}_\phi$$

within the conductor. Find the corresponding current density.

(b) If $I = 3$ A and $a = 2$ cm in part (a), find $\mathbf{H}$ at (0, 1 cm, 0) and (0, 4 cm, 0).

7.22 An infinitely long cylindrical conductor of radius a is placed along the z-axis. If the current density is $\mathbf{J} = \frac{J_o}{\rho}\mathbf{a}_z$, where J_0 is constant, find $\mathbf{H}$ everywhere.

7.23 Let $\mathbf{H} = k_o\left(\frac{\rho}{a}\right)\mathbf{a}_\phi$, $\rho < a$, where k_o is a constant, (a) Find $\mathbf{J}$ for $\rho < a$. (b) Find $\mathbf{H}$ for $\rho > a$.

7.24 Let $\mathbf{H} = y^2\mathbf{a}_x + x^2\mathbf{a}_y$ A/m. Find $\mathbf{J}$ at $(1, -4, 7)$.

7.25 Assume a conductor, $\mathbf{H} = 10^3\rho^2\mathbf{a}_\phi$ A/m. (a) Find $\mathbf{J}$. (b) Calculate the current through the surface $0 < \rho < 2, 0 < \phi < 2\pi, z = 0$.

7.26 Consider the two-wire transmission line whose cross section is illustrated in Figure 7.37. Each wire is of radius 2 cm, and the wires are separated 10 cm. The wire centered at (0, 0) carries a current of 5 A while the other centered at (10 cm, 0) carries the return current. Find $\mathbf{H}$ at

(a) (5 cm, 0)

(b) (10 cm, 5 cm)

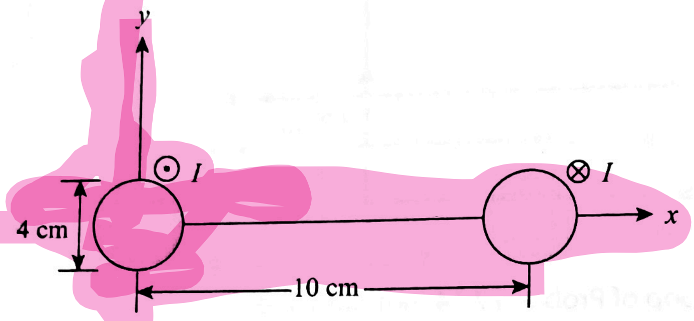

Figure 7.37 Two-wire line of Problem 7.26.

7.27 An infinitely long filamentary wire carries a current of 2 A along the z-axis in the $+z$-direction. Calculate the following:

(a) $\mathbf{B}$ at $(-3, 4, 7)$

(b) The flux through the square loop described by $2 \le \rho \le 6$, $0 \le z \le 4$, $\phi = 90°$.

7.28 An electron beam forms a current of density $\mathbf{J} = \begin{cases} J_0(1 - \rho^2/a^2)\mathbf{a}_z, & \rho < a \\ 0, & \rho > a \end{cases}$

(a) Determine the total current.

(b) Find the magnetic field intensity everywhere.

Section 7.5—Magnetic Flux Density

7.29 Determine the magnetic flux through a rectangular loop ($a \times b$) due to an infinitely long conductor carrying current I as shown in Figure 7.38. The loop and the straight conductors are separated by distance d.

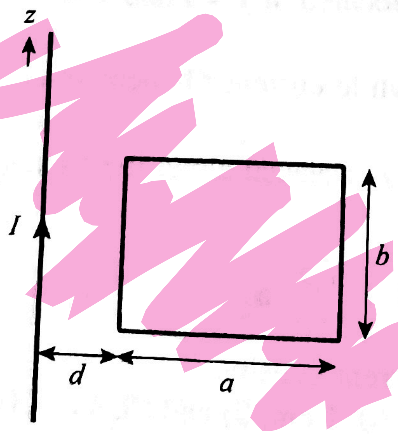

Figure 7.38 For Problem 7.29.

7.30 A semicircular loop of radius a in free space carries a current I. Determine the magnetic flux density at the center of the loop.

7.31 In free space, the magnetic flux density is

$$\mathbf{B} = y^2\mathbf{a}_x + z^2\mathbf{a}_y + x^2\mathbf{a}_z \text{ Wb/m}^2$$

(a) Show that $\mathbf{B}$ is a magnetic field
(b) Find the magnetic flux through $x = 1, 0 < y < 1, 1 < z < 4$.
(c) Calculate $\mathbf{J}$.

*7.32 A brass ring with triangular cross section encircles a very long straight wire concentrically as in Figure 7.39. If the wire carries a current I, show that the total number of magnetic flux lines in the ring is

$$\Psi = \frac{\mu_o I h}{2\pi b}\left[b - a \ln \frac{a+b}{b}\right]$$

Calculate Ψ if $a = 30$ cm, $b = 10$ cm, $h = 5$ cm, and $I = 10$ A.

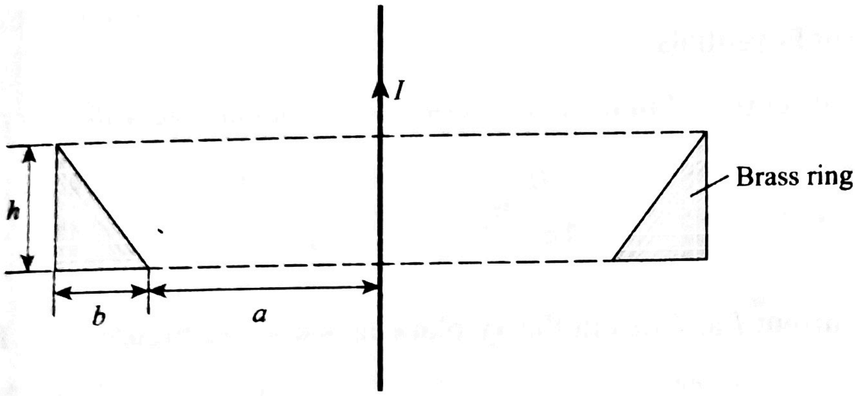

Figure 7.39 Cross section of a brass ring enclosing a long straight wire; for Problem 7.32.

7.33 The electric motor shown in Figure 7.40 has field

$$\mathbf{H} = \frac{10^6}{\rho}\sin 2\phi\, \mathbf{a}_\rho \text{ A/m}$$

Calculate the flux per pole passing through the air gap if the axial length of the pole is 20 cm.

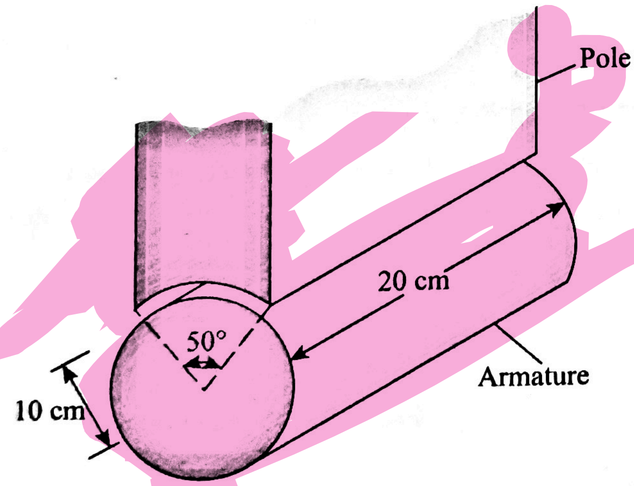

Figure 7.40 Electric motor pole of Problem 7.33.

7.34 In free space, $\mathbf{B} = \frac{20}{\rho}\sin^2\phi\,\mathbf{a}_z$ Wb/m^2. Determine the magnetic flux crossing the strip $z = 0$, $1 < \rho < 2$ m, $0 < \phi < \pi/4$.

7.35 If $\mathbf{B} = \frac{2}{r^3}\cos\theta\,\mathbf{a}_r + \frac{1}{r^3}\sin\theta\,\mathbf{a}_\theta$ Wb/m^2, find the magnetic flux through the spherical cap $r = 1$, $\theta < \pi/3$.

7.36 In a hydrogen atom, an electron revolves at velocity 2.2×10^6 m/s. Calculate the magnetic flux density at the center of the electron's orbit. Assume that the radius of the orbit is $R = 5.3 \times 10^{-11}$ m.

Section 7.6—Maxwell's Equations

7.37 Consider the following arbitrary fields. Find out which of them can possibly represent an electrostatic or magnetostatic field in free space.

(a) $\mathbf{A} = y \cos ax\mathbf{a}_x + (y + e^{-x})\mathbf{a}_z$

(b) $\mathbf{B} = \dfrac{20}{\rho}\mathbf{a}_\rho$

(c) $\mathbf{C} = r^2 \sin\theta\,\mathbf{a}_\phi$

7.38 Reconsider Problem 7.37 for the following fields,

(a) $\mathbf{D} = y^2 z\mathbf{a}_x + 2(x + 1)yz\mathbf{a}_y - (x+1)z^2\mathbf{a}_z$

(b) $\mathbf{E} = \dfrac{(z+1)}{\rho}\cos\phi\,\mathbf{a}_\rho + \dfrac{\sin\phi}{\rho}\mathbf{a}_z$

(c) $\mathbf{F} = \dfrac{1}{r^2}(2\cos\theta\,\mathbf{a}_r + \sin\theta\,\mathbf{a}_\theta)$

Section 7.7—Magnetic Scalar and Vector Potentials

7.39 A current element of length L carries current I in the z direction. Show that at a very distant point,

$$\mathbf{A} = \frac{\mu_o IL}{4\pi r}\mathbf{a}_z$$

Find **B**.

7.40 A square loop of side $2a$ carries current I and lies in the xy-plane as shown in Figure 7.41. Find **A** at point $P(2a, 0, 0)$.

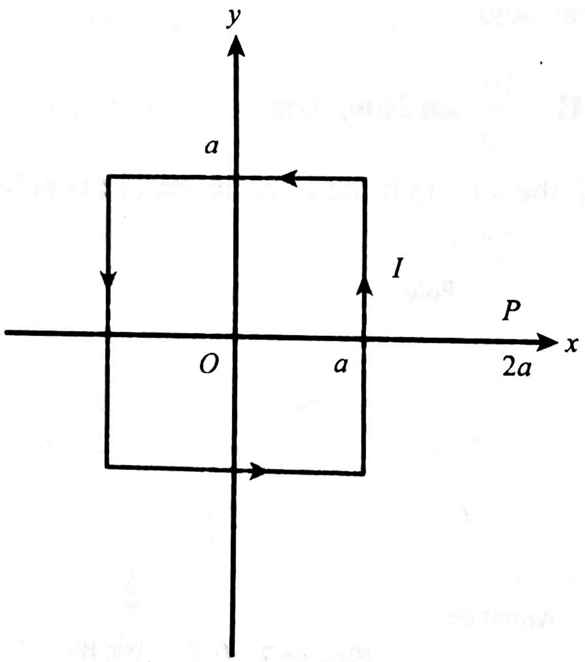

Figure 7.41 For Problem 7.40.

7.41 Find the **B** field corresponding to the magnetic vector potential

$$\mathbf{A} = \sin\frac{\pi x}{2}\cos\frac{\pi y}{2}\mathbf{a}_z$$

7.42 In a certain region ($\mu = \mu_o$), the magnetic vector potential is

$$\mathbf{A} = -\mu_o k(x^2 + y^2)\mathbf{a}_z \text{ Wb/m}$$

where k is a constant. Determine the magnetic field intensity **H**.

7.43 Given the following two magnetic vector potentials

$$\mathbf{A}_1 = (\sin x + x \sin y)\mathbf{a}_y$$

$$\mathbf{A}_2 = \cos y\mathbf{a}_x + \sin x\mathbf{a}_y$$

show that they give the same magnetic flux density $\mathbf{B}$. Show that $\mathbf{B}$ is solenoidal.

7.44 The magnetic vector potential of a current distribution in free space is given by

$$\mathbf{A} = 15e^{-\rho} \sin\phi \, \mathbf{a}_z \text{ Wb/m}$$

Find $\mathbf{H}$ at $(3, \pi/4, -10)$. Calculate the flux through $\rho = 5, 0 \le \phi \le \pi/2, 0 \le z \le 10$.

7.45 Given that $\mathbf{A} = \dfrac{10}{r}\sin\theta\,\mathbf{a}_\phi$ Wb/m find $\mathbf{H}$ at point $(4, 60°, 30°)$.

7.46 An infinitely long conductor of radius a carries a uniform current with $\mathbf{J} = J_o\mathbf{a}_z$. Show that the magnetic vector potential for $\rho < a$ is

$$\mathbf{A} = -\frac{1}{4}\mu_o J_o \rho^2 \mathbf{a}_z$$

7.47 In free space, $A = 10 \sin \pi y\mathbf{a}_x + (4 + \cos \pi x)\mathbf{a}_z$ Wb/m. Find $\mathbf{H}$ and $\mathbf{J}$.

7.48 The magnetic vector potential at a distant point from a small circular loop is given by

$$\mathbf{A} = \frac{A_o}{r^2}\sin\theta\,\mathbf{a}_\phi \text{ Wb/m}$$

where A_o is a constant. Determine the magnetic flux density $\mathbf{B}$.

7.49 The magnetic field intensity in a certain conducting medium is

$$\mathbf{H} = xy^2\mathbf{a}_x + x^2z\mathbf{a}_y - y^2z\mathbf{a}_z \text{ A/m}$$

(a) Calculate the current density at point $P(2, -1, 3)$.

(b) What is $\dfrac{\partial \rho_v}{\partial t}$ at P?

7.50 Let $\mathbf{A} = 10\rho^2\mathbf{a}_z\,\mu$ Wb/m.

(a) Find $\mathbf{H}$ and $\mathbf{J}$.

(b) Determine the total current crossing the surface $z = 1, 0 \le \rho \le 2, 0 \le \phi \le 2\pi$.

7.51 Prove that the magnetic scalar potential at $(0, 0, z)$ due to a circular loop of radius a shown in Figure 7.8(a) is

$$V_m = \frac{I}{2}\left[1 - \frac{z}{[z^2 + a^2]^{1/2}}\right]$$

***7.52** If $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ and $R = |\mathbf{R}|$, show that

$$\nabla\frac{1}{R} = -\nabla'\frac{1}{R} = -\frac{\mathbf{R}}{R^3}$$

where ∇ and ∇' are del operators with respect to (x, y, z) and (x', y', z'), respectively.